

Technology Stack

Date	02 July 2026
Team ID	SWTID-2026-9778
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Python and Scikit-learn enable accurate machine learning predictions, Flask efficiently connects the web application with the prediction model, MySQL securely stores applicant data, and modern frontend technologies create an intuitive user interface. Together, these technologies ensure the application is scalable, secure, maintainable, reliable, and capable of delivering fast and accurate credit card approval decisions..

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	<i>HTML5, CSS3, JavaScript, Bootstrap 5</i>	<i>Used to build a responsive, user-friendly, and mobile-compatible interface for applicants and administrators. Bootstrap enables faster UI development with reusable components.</i>

2	Backend / Server-Side	<i>Python, Flask</i>	<i>Python provides excellent support for Machine Learning libraries, while Flask is lightweight, fast, and easy to integrate with ML models and REST APIs.</i>
3	Database / Data Storage	<i>MySQL</i>	<i>Stores applicant details, application history, prediction results, and user credentials securely. MySQL offers strong relational data management and high reliability.</i>
4	Cloud / Hosting / Deployment	<i>Render / Railway / AWS EC2 (or PythonAnywhere for academic deployment)</i>	<i>Provides reliable cloud hosting with easy deployment, scalability, automatic updates, and high availability for the web application.</i>
5	Version Control & CI/CD	<i>Git, GitHub</i>	<i>Enables source code management, version tracking, collaboration among team members, backup, and continuous integration using GitHub Actions (optional).</i>

6	Third-Party APIs / Other Tools	<i>Scikit-learn, Pandas, NumPy, Matplotlib.</i>	<i>Scikit-learn is used to train the Credit Card Approval Prediction model. Pandas and NumPy handle data preprocessing, Matplotlib supports data visualization, Joblib saves the trained model, and SMTP Email API sends application status notifications to applicants</i>
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